

Multi-Modal Histopathological Breast Cancer Classification via Hybrid EfficientNet and Morphological Feature Fusion with Attention Mechanisms

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Abstract—Breast cancer remains the most commonly diagnosed cancer and a of the leading causes of cancer mortality among women worldwide, which makes an accurate histopathological classification essential for early diagnosis and treatment planning. Convolutional neural network (CNN) based approaches have advanced automated histopathology analysis but typically operate as “black box” visual feature extractors that ignore the quantitative morphological criteria – such as nuclear shape, size, and boundary regularity – that pathologists routinely use during manual diagnosis. We propose a hybrid classification framework that combines visual embeddings of EfficientNet-B0 with seven interpretable morphological descriptors (area, perimeter, eccentricity, solidity, extent, mean intensity, and compactness) computed from Otsu-segmented Macenko-normalized tissue regions. Three fusion strategies are designed and benchmarked in the BreakHis histopathological image dataset across four magnification levels (40X–400X) and eight benign/malignant subtypes. The attention-weighted fusion strategy achieves the best performance (98.02% accuracy, 97.8% precision, 98.3% recall, 98.0% F1-score), significantly outperforming a CNN-only baseline (75.26%) and prior state-of-the-art methods reported on BreakHis. An ablation study confirms that stain normalization and morphological feature fusion each contribute significant, statistically validated (paired t -test, $p < 0.05$ after Bonferroni correction) accuracy gains. Beyond raw performance, the attention weights over the fused 519-dimensional feature vector provide a sample-level explanation of which morphological and visual cues drove each prediction, offering a practical route toward clinically trustworthy, deployable computer-aided diagnosis.

Index Terms—Breast cancer classification, hybrid CNN-morphological features, attention mechanisms, multi-modal fusion, histopathological images, Macenko stain normalization, deep learning, transfer learning, interpretability

I. Introduction

Breast cancer is the second most common cancer worldwide and remains a leading cause of cancer-related death among women, with roughly 2.3 million new cases and over 685,000 deaths reported annually [1]. Although the five-year survival rate following early detection exceeds 90%, definitive diagnosis still depends on histopathological examination of tissue biopsies by trained pathologists [2]. This manual process is slow, expertise-dependent, and subject to inter-observer variability, with reported diagnostic agreement in difficult cases falling to 75–85% [3]. These limitations motivate automated,

objective computational tools that can serve as a reliable second opinion, particularly in resource-constrained healthcare settings that lack ready access to expert pathologists.

Deep convolutional neural networks (CNNs) – including AlexNet, VGG, ResNet, DenseNet, and EfficientNet – have substantially improved automated histopathology classification by learning hierarchical visual representations directly from pixel data [4], [5]. However, CNN-only pipelines suffer from two persistent limitations in the medical domain. First, they function largely as black boxes, offering little insight into *why* a particular classification was made, which limits clinical trust and regulatory acceptance [6]. Second, they do not explicitly encode the quantitative morphological criteria – cell and nuclear shape, size, boundary regularity, and tissue organization – that pathologists are trained to assess [7]. Compounding these issues, histopathological images exhibit substantial color and intensity variation arising from differences in staining protocol, reagent concentration, and scanner characteristics, which degrades cross-laboratory generalization unless explicitly corrected [8], [9].

This gap between what CNNs learn and what pathologists actually reason about has direct clinical consequences. Regulatory frameworks for software-as-a-medical-device increasingly require that automated decision support provide a rationale that a clinician can audit and cross-check against established diagnostic criteria, not merely a confidence score [6], [10]. A purely visual CNN embedding cannot be decomposed into the nuclear and tissue-level attributes – shape irregularity, boundary complexity, size variation – that already form the vocabulary of manual grading, which makes it difficult for a pathologist to verify *agreement* or diagnose *disagreement* with the model on a given slide. At the same time, discarding these descriptors entirely, as purely handcrafted-feature pipelines historically did, sacrifices the discriminative power of learned visual representations. This motivates a hybrid design in which interpretable morphological measurements are not used as a replacement for CNN features but are fused with them in a way that preserves each modality’s per-sample contribution, so that predictions remain both accurate and traceable to clinically meaningful evidence. Achieving this requires deliberately ad-

addressing the stain-variability confound described above, since morphological descriptors computed on inconsistently stained tissue are themselves unreliable, and a fusion mechanism that can adaptively weight visual versus morphological evidence rather than treating them as fixed, equally-weighted inputs.

In this work, we address these limitations with a hybrid classification framework that fuses deep CNN features with a compact, clinically motivated set of morphological descriptors. Our specific contributions are:

- A hybrid breast cancer classification pipeline that fuses EfficientNet-B0 visual embeddings with seven quantitative morphological descriptors, improving both accuracy and interpretability on histopathology images.
- Integration of Macenko stain normalization [8] and a robust preprocessing pipeline to reduce stain variability across slides and scanners.
- Design and evaluation of three multi-modal fusion strategies – concatenation, attention-weighted fusion, and separate-then-combine – with attention-weighted fusion shown to perform best.
- A comprehensive experimental protocol (five-fold cross-validation, ablation study, paired significance testing) demonstrating that the reported gains are statistically robust rather than artifacts of a single train/test split.

II. Related Work

Breast cancer histopathology classification has progressively evolved from handcrafted image descriptors and conventional machine-learning classifiers toward deep convolutional neural networks (CNNs), transfer learning, hybrid feature representations, attention mechanisms, and transformer-based architectures. The BreakHis dataset has been particularly important in this development because it contains histopathological images acquired at four magnification levels (40X, 100X, 200X, and 400X) and includes both benign and malignant breast-tumor subtypes. Earlier studies established the difficulty of this multi-class and multi-magnification problem, while subsequent work has increasingly focused on improving feature representation, robustness, and computational efficiency.

II-A Deep Learning and Neural Architecture Search

By 2021, research had moved beyond manually designed CNN architectures toward automated network design and hybrid learning pipelines. Oyelade and Ezugwu [11] proposed a bioinspired neural architecture search (NAS) framework for breast cancer histopathology classification. Their approach employed a block-based stochastic categorical-to-binary search-space initialization together with an Ebola Optimization Search Algorithm to identify suitable CNN architectures. Experiments were conducted using both BACH and BreakHis, demonstrating the potential of automated architecture design for histopathological image analysis. This work illustrates the increasing interest in optimizing CNN structure specifically for the characteristics of medical images rather than relying exclusively on manually selected architectures.

Liew et al. [12] investigated a complementary hybrid strategy by combining transfer-learned DenseNet201 features with XGBoost for classification of BreakHis images. Their Deep Learning and eXtreme Gradient Boosting (DLXGB) framework addressed both binary benign/malignant classification and the eight-class histological problem and reported 97% accuracy for both settings. The study demonstrates that learned deep representations can be effectively combined with a conventional machine-learning classifier rather than relying exclusively on a neural-network classification head.

In 2022, Chattopadhyay et al. [13] introduced MTRRE-Net, a Multi-scale Twofold Residual Recurrent Network for breast cancer histopathology classification. The model combines multi-scale feature learning, dual residual blocks, and recurrent connections to improve feature propagation and address degradation and vanishing-gradient problems in deeper networks. Evaluated on BreakHis, MTRRE-Net achieved accuracies of 97.12%, 95.22%, 96.85%, and 97.81% at 40X, 100X, 200X, and 400X magnifications, respectively. This study highlights the importance of multi-scale representations for handling the substantial visual variation caused by different acquisition magnifications.

In contrast to purely learned representations, Joseph et al. [14] investigated the use of explicit handcrafted descriptors combined with a deep neural-network classifier. Their method extracted Hu moments, Haralick texture features, and color histograms from BreakHis images and used the resulting descriptors as inputs to a dense neural network. The method achieved 97.87%, 97.60%, 96.10%, and 96.84% accuracy at 40X, 100X, 200X, and 400X, respectively. This result is particularly relevant to the present work because it demonstrates that explicit image descriptors can provide useful complementary information for histopathological classification.

II-B Transfer Learning and Hybrid Deep Features

Recent work has increasingly investigated transfer learning and combinations of deep representations with conventional classifiers. Maleki et al. [15] proposed a DenseNet201-XGBoost framework for breast cancer diagnosis from histopathology images. Their study evaluated multiple combinations of pretrained feature extractors and gradient-boosting algorithms and reported improved performance on the BreakHis dataset. The work reinforces the observation that feature extraction and classification can be optimized as separate components, rather than requiring a completely end-to-end neural architecture.

Toma et al. [16] further investigated transfer learning using a broad collection of pretrained architectures, including ResNet, ResNeXt, SENet, Dual Path Networks, DenseNet, NASNet, and Wide ResNet. Using BreakHis, their experiments demonstrated very high classification accuracies for several architectures, including 99.8% with ResNeXt-50 and 99.725% with DenseNet-169. These results demonstrate the effectiveness of pretrained CNN representations for this dataset, but they also indicate that raw accuracy comparisons must be interpreted carefully because differences in preprocessing, class

formulation, train/test partitioning, and evaluation protocol can substantially influence reported results.

In 2024, DeVoe et al. [17] evaluated MobileNetV3 for breast histopathology classification on BreakHis. Their results demonstrated that lightweight architectures can achieve strong classification performance while reducing the computational complexity associated with larger CNNs. This direction is important for practical computer-aided diagnosis systems in which inference cost and deployment constraints are relevant in addition to classification accuracy.

Potsangbam and Devi [18] investigated transfer learning with DenseNet121 for binary breast cancer classification using BreakHis. Their method used a pretrained DenseNet121 feature extractor followed by fully connected layers and reported a maximum accuracy of 96.09% at 100X magnification. The study further confirms the value of pretrained CNNs for histopathological image analysis and provides a useful comparison with other transfer-learning architectures.

II-C Attention and Transformer-Based Histopathology Classification

The recent literature has increasingly shifted toward attention-based and transformer architectures to capture long-range spatial dependencies and more selectively represent diagnostically relevant image regions. Yan et al. [19] proposed DWNAT-Net, which combines the Discrete Wavelet Transform (DWT) with a Neighborhood Attention Transformer (NAT). The DWT component captures frequency-domain information while NAT models local token relationships with reduced attention complexity. Evaluated on BreakHis and BACH, the model achieved 99.66% recognition accuracy on BreakHis, demonstrating the potential of combining spatial and frequency-domain representations.

Rafiq et al. [20] proposed an enhanced DenseNet121-based framework for multi-class breast cancer classification using histopathological images. The study evaluated the model on BreakHis and reported 98.50% binary classification accuracy and 92.50% multi-class accuracy. The results demonstrate that recent CNN-based systems continue to provide competitive performance, particularly when preprocessing, augmentation, and feature-learning strategies are carefully designed.

Transformer-based approaches have continued to develop rapidly in 2026. Balaha et al. [21] introduced a computer-aided diagnosis framework based on two Vision Transformer models for multistage histopathology-based breast cancer classification. Using BreakHis, their framework achieved 98.99% accuracy, 98.13% precision, and 98.00% recall. This work illustrates the increasing use of global self-attention for modeling complex tissue structures in breast histopathology.

Uddin and Adnan [22] proposed TransBreast-Net, an ensemble of Vision Transformer architectures including CaiT, DeiT, and Swin Transformer. The framework combines complementary transformer representations and incorporates Grad-CAM visualizations to improve interpretability. On BreakHis, the method achieved 99.35% binary classification accuracy and 97% multi-class accuracy. Although these results demonstrate

the strong discriminative capability of transformer ensembles, the approach remains dependent primarily on learned visual representations rather than explicitly quantified morphological descriptors.

II-D Research Gap and Motivation for the Proposed Framework

The progression of the literature reveals two complementary directions. First, deep CNN and transformer models provide highly discriminative learned representations and have produced increasingly high classification accuracy on BreakHis. However, these learned representations are difficult to interpret directly in terms of explicit pathological characteristics such as tissue area, boundary regularity, shape, compactness, and related morphological properties. Second, handcrafted-feature approaches provide more explicit and interpretable measurements, but their representational capacity is generally more limited than that of modern deep networks.

Recent hybrid methods have demonstrated that combining learned and engineered representations can be effective; however, most existing studies focus on either combining deep features with conventional classifiers or concatenating multiple feature types. Comparatively less attention has been given to systematically evaluating whether an adaptive fusion mechanism can determine the relative contribution of visual and morphological information for individual histopathology images.

The proposed framework addresses this gap by combining a pretrained EfficientNet-B0 visual representation with seven quantitative morphological descriptors: area, perimeter, eccentricity, solidity, extent, mean intensity, and compactness. The two modalities are evaluated using three controlled fusion strategies: direct concatenation, attention-weighted fusion, and separate-then-combine prediction. The attention mechanism allows the contribution of the 512-dimensional deep visual representation and the seven-dimensional morphological representation to be weighted adaptively rather than assuming that both modalities are equally informative for every sample.

The proposed study further differs from previous work in its experimental design. All three fusion strategies are evaluated under the same preprocessing and training protocol, while five-fold cross-validation, ablation experiments, and paired statistical significance testing are used to assess the stability of the observed improvements. Thus, the contribution of the present work is not simply another CNN architecture, but a controlled investigation of how interpretable morphological evidence can complement an efficient deep visual representation through adaptive multimodal fusion. ““

III. Methodology

Figure 1 shows the end-to-end pipeline. An input hematoxylin-and-eosin (H&E) image is preprocessed and passed through two parallel branches: (i) a CNN branch that produces a compact visual embedding, and (ii) a morphology branch that produces seven quantitative tissue descriptors from a segmented binary mask. The two feature sets are combined

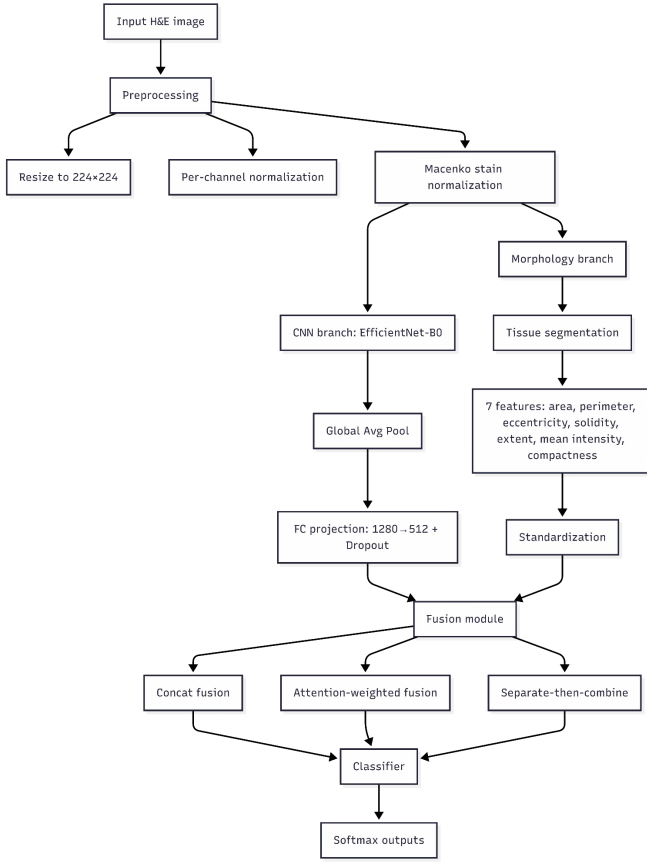


Fig. 1: End-to-end block diagram of the proposed hybrid CNN + morphological-feature classification pipeline. Input H&E images are preprocessed (resize to 224×224 , per-channel normalization, Macenko stain normalization) and processed by two parallel branches: an EfficientNet-B0 CNN branch producing a 512-dimensional visual embedding, and a morphology branch extracting seven tissue descriptors via binary segmentation. Outputs are fused (concatenation, attention-weighted, or separate-then-combine) before softmax classification.

by one of three fusion strategies before a final softmax classification into eight histological subtypes.

III-A Dataset

We use the publicly available BreakHis dataset [2], comprising 7,909 microscopic images from 82 patients, digitized at four magnifications (40X, 100X, 200X, 400X) as 700×460 RGB PNG images. Images are labeled as benign (2,480 images: Adenosis, Fibroadenoma, Phyllodes Tumor, Tubular Adenoma) or malignant (5,429 images: Ductal, Lobular, Mucinous, and Papillary Carcinoma). A stratified 80:20 train/test split is applied, preserving the proportion of each subtype and magnification level in both partitions. Fig. 2 summarizes the resulting per-magnification sample counts and the benign/malignant class balance.

Representative examples of benign and malignant histopathological images at the four available magnification

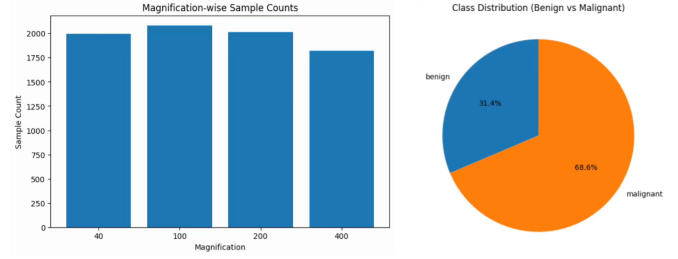


Fig. 2: BreakHis dataset statistics. Left: sample counts per magnification level. Right: class distribution (31.4% benign vs. 68.6% malignant), motivating the use of stratified sampling.

Representative BreakHis Histopathological Images

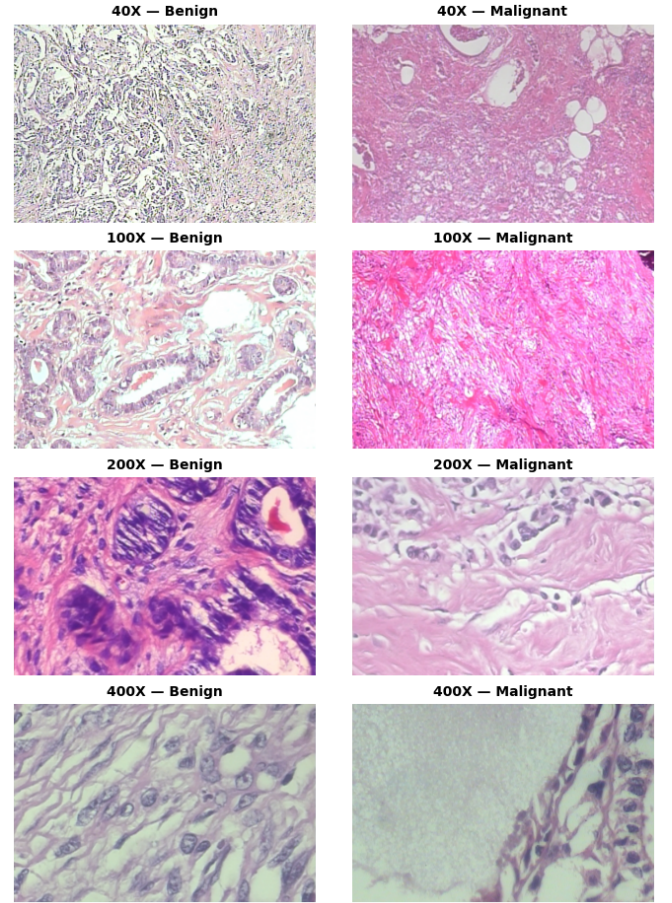


Fig. 3: Representative BreakHis histopathological images at four magnification levels. Each row shows benign and malignant examples at 40X, 100X, 200X, and 400X magnifications, respectively.

levels are shown in Fig. 3. The examples illustrate the variation in tissue appearance and cellular morphology across 40X, 100X, 200X, and 400X magnifications.

III-B Preprocessing

Images are first stain-normalized using the Macenko method [8]. The method decomposes each image into optical density space using the Beer–Lambert law:

$$\text{OD} = -\log\left(\frac{I + \epsilon}{I_0}\right), \quad (1)$$

where I is the observed intensity, I_0 is the incident light intensity, and ϵ is a small constant used to avoid numerical instability. Eigen-decomposition of the optical-density covariance matrix identifies the dominant hematoxylin and eosin stain vectors. The stain concentrations are then re-normalized to a reference color template according to

$$\begin{aligned} \text{OD} &\approx WH, \\ \text{OD}_{\text{norm}} &= W_{\text{ref}} H_{\text{scaled}}, \\ I_{\text{norm}} &= I_0 \exp(-\text{OD}_{\text{norm}}). \end{aligned} \quad (2)$$

The normalized images are subsequently resized to 224×224 pixels using bicubic interpolation and normalized using the standard ImageNet channel statistics, $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$. During training, data augmentation is applied dynamically and balanced across classes and magnification levels. The augmentation pipeline includes random rotation within $\pm 90^\circ$, horizontal and vertical flipping, scaling in the range $0.9\text{--}1.1\times$, and brightness and contrast jitter of $\pm 10\%$.

III-C Morphological Feature Extraction

Tissue regions are segmented from the stain-normalized image using Otsu’s automatic thresholding, which selects the threshold t^* that maximizes inter-class variance between tissue and background:

$$t^* = \arg \max_t \sigma_B^2(t), \quad \sigma_B^2(t) = \omega_0(t) \omega_1(t) (\mu_0(t) - \mu_1(t))^2. \quad (3)$$

Morphological closing followed by opening (3×3 structuring element) removes noise while preserving genuine tissue boundaries. From the resulting binary mask R , we compute seven descriptors that mirror criteria used by pathologists during manual grading:

$$\text{Area: } A = |R| \quad (4)$$

$$\text{Perimeter: } P = \text{length}(\partial R) \quad (5)$$

$$\text{Eccentricity: } e = \sqrt{1 - b^2/a^2} \quad (6)$$

$$\text{Solidity: } = A/A_{\text{convex hull}} \quad (7)$$

$$\text{Extent: } = A/A_{\text{bbox}} \quad (8)$$

$$\text{Mean intensity: } \mu_R = \frac{1}{A} \sum_{(x,y) \in R} I_g(x,y) \quad (9)$$

$$\text{Compactness: } C = P^2/(4\pi A) \quad (10)$$

All seven features are standardized (zero mean, unit variance) using statistics computed on the training set only, to avoid data leakage and to prevent features with larger numerical ranges (e.g., area, perimeter) from dominating the fused representation.

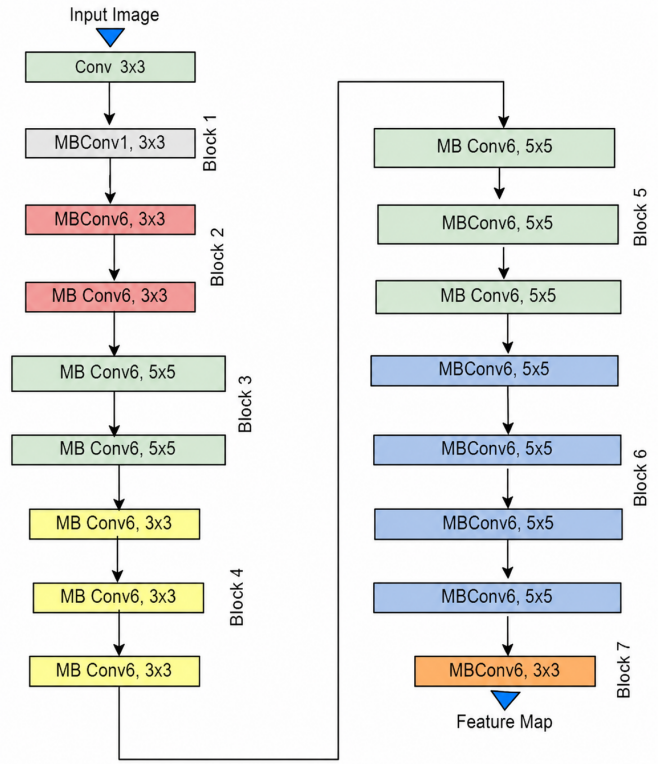


Fig. 4: Schematic of the EfficientNet-B0 backbone. Seven MBConv stages with progressively increasing resolution, channel width, and depth; each block uses depthwise separable convolution and squeeze-and-excitation attention. Global average pooling yields a 1280-dimensional embedding, subsequently projected to 512 dimensions.

III-D CNN Backbone

We use EfficientNet-B0 [5], pretrained on ImageNet, as the visual feature extractor. EfficientNet’s compound scaling relates depth, width, and resolution scaling coefficients α, β, γ to a single scale factor ϕ :

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi. \quad (11)$$

Global average pooling converts the final 1,280-dimensional feature map into a fixed-length vector, which is projected to 512 dimensions via a fully connected layer with ReLU activation and dropout. Early layers are frozen to retain generic visual primitives, while deeper layers are fine-tuned to learn domain-specific histopathological patterns. Fig. 4 illustrates the seven-stage MBConv backbone used for this branch.

III-E Multi-Modal Fusion Strategies

We investigate three strategies for combining the 512-dimensional CNN embedding f_{cnn} with the 7-dimensional standardized morphological vector f_{morph} .

Concatenation fusion. The simplest strategy concatenates the two feature vectors into a single 519-dimensional representation,

$$f = [f_{\text{cnn}}; f_{\text{morph}}], \quad (12)$$

which is passed to a three-layer MLP classifier ($519 \rightarrow 256 \rightarrow 128 \rightarrow \text{num_classes}$) with batch normalization, ReLU, and dropout (0.4 / 0.2).

Attention-weighted fusion. To let the model adaptively re-weight the contribution of each feature dimension per sample, an attention sub-network computes a gate vector $a \in (0, 1)^{519}$ over the concatenated features:

$$a = \sigma(W_2 \text{ReLU}(W_1 f + b_1) + b_2), \quad (13)$$

$$\tilde{f} = a \odot f, \quad (14)$$

where σ is the sigmoid function and $\odot$ is the element-wise product. The gated vector $\tilde{f}$ is passed to the same MLP classifier architecture as the concatenation baseline. This mechanism introduces a modest 133,375 additional parameters and provides (i) adaptive modality balancing that prevents CNN-feature dominance, (ii) input-conditioned specialization (e.g., emphasizing solidity for malignancy assessment or eccentricity for ductal patterns), and (iii) an interpretable, per-sample map of feature importance.

Separate-then-combine fusion. The most complex strategy trains two independent classifiers – a CNN-specific MLP ($512 \rightarrow 256 \rightarrow \text{num_classes}$) and a morphology-specific MLP ($7 \rightarrow 64 \rightarrow \text{num_classes}$) – and combines their softmax outputs p_c and p_m using learned weights:

$$p = \lambda p_c + (1 - \lambda) p_m, \quad \lambda \in [0, 1]. \quad (15)$$

III-F Training and Optimization

All models are trained with cross-entropy loss,

$$\mathcal{L} = - \sum_{i=1}^C y_i \log(p_i), \quad (16)$$

using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$), an initial learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , and a StepLR schedule ($\gamma = 0.1$ every 15 epochs). Training runs for up to 50 epochs with early stopping (patience = 10) based on validation accuracy, using a batch size of 32. Table II summarizes the full hyperparameter configuration.

All experiments were implemented in Python 3.10 using PyTorch 2.x, torchvision, and scikit-learn for evaluation metrics and statistical testing, and were run on the Google Colab platform with dual NVIDIA Tesla T4 GPUs (16 GB VRAM each, CUDA 12.x) *[RAM, CPU, and OS build as allocated by the Colab runtime instance – please confirm/insert your exact session specs]*. A full 50-epoch run of the attention-weighted model took approximately 6 hours. Table I summarizes the software and hardware environment.

IV. Experiments and Results

IV-A Classification Performance

Table III compares the three fusion strategies on the held-out BreakHis test set. Attention-weighted fusion achieves the best performance across all four metrics, which is also reflected in the confusion matrix and multi-class ROC curves of Fig. 5.

TABLE I: Experimental hardware and software environment.

Component	Specification
Platform	Google Colab
GPU	2× NVIDIA Tesla T4 (16 GB VRAM each)
CUDA / cuDNN	CUDA 12.x / cuDNN 8.x
RAM	<i>[confirm Colab runtime RAM]</i>
CPU	<i>[confirm Colab runtime vCPUs]</i>
OS	Ubuntu (Colab default image)
Language	Python 3.10
Deep learning framework	PyTorch 2.x, torchvision
Supporting libraries	NumPy, scikit-learn, scikit-image, OpenCV, SciPy

TABLE II: Hyperparameter configuration.

Hyperparameter	Value
Backbone	EfficientNet-B0 (ImageNet-pretrained)
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Initial learning rate	1×10^{-4}
LR scheduler	StepLR ($\gamma=0.1$, step = 15 epochs)
Weight decay	1×10^{-4}
Batch size	32
Epochs	50 (early stopping, patience = 10)
Dropout	0.4 (FC1), 0.2 (FC2)
CNN feature dim.	1280 $\rightarrow$ 512
Morphological features	7 (z-score normalized)
Fused feature dim.	519
Train / test split	80:20 (stratified)
Cross-validation	5-fold stratified

Table IV reports accuracy of the attention-weighted model at each magnification level. Performance is stable across scales (97.84–98.12%), with 100X achieving the best result, plausibly reflecting a favorable balance between cellular detail and tissue-level context at that intermediate scale.

IV-B Ablation Study

Table V reports the contribution of each pipeline component. Removing Macenko stain normalization causes a 15.66 percentage-point accuracy drop, confirming its critical role in handling cross-slide color variability. Adding morphological features to the CNN-only baseline increases accuracy by 22.39 percentage points, and attention-weighted fusion adds a further 0.37 points over simple concatenation. Fig. 6 shows the corresponding training accuracy and loss curves for each fusion strategy.

IV-C Cross-Validation and Statistical Significance

Stratified five-fold cross-validation (Table VI) shows low variance across folds ($98.00\% \pm 0.20\%$ accuracy), confirming stable generalization without significant overfitting. Paired t -tests with Bonferroni correction (Table VII) confirm that all improvements are statistically significant ($p < 0.05$, corrected $\alpha = 0.0125$); the comparison against the CNN-only baseline is the strongest ($t = 31.42$, $p < 0.001$), and attention-weighted fusion significantly outperforms both simpler fusion strategies.

IV-D Comparison with Prior Work

Table VIII summarizes representative recent studies on the BreakHis dataset from 2021 to 2026. These studies employ a

TABLE III: Classification performance of the three fusion strategies on the BreakHis test set (weighted average across eight subtypes).

Strategy	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
Concatenation	97.65	97.2	97.8	97.5
Attention-weighted	98.02	97.8	98.3	98.0
Separate-then-combine	97.45	96.9	97.7	97.3

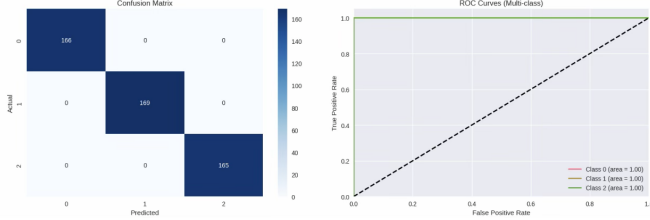


Fig. 5: (Left) Confusion matrix and (right) multi-class ROC curves for the attention-weighted fusion model on the held-out test set.

range of approaches, including neural architecture search, hybrid CNN–XGBoost models, handcrafted descriptors, transfer learning, lightweight CNNs, and transformer-based architectures.

The proposed framework achieves 98.02% accuracy for the eight-class classification task. This result is competitive with recent BreakHis-based methods, although some studies report higher numerical accuracies under different experimental settings. For example, Toma et al. [16] reported accuracies up to 99.8%, while Yan et al. [19] reported 99.66% using a discrete wavelet transform and Neighborhood Attention Transformer. Balaha et al. [21] also reported 98.99% using a Vision Transformer framework. These results demonstrate the strong performance of increasingly sophisticated deep learning and transformer-based architectures.

However, direct numerical comparison should be interpreted cautiously because the studies differ in classification formulation, magnification-specific evaluation, preprocessing, data partitioning, and reported metrics. In particular, some reported results correspond to binary classification rather than the eight-class classification task considered in this work. The proposed framework therefore does not claim to achieve the highest reported accuracy on BreakHis. Instead, its primary contribution is the integration of EfficientNet-B0 visual features with explicit morphological descriptors through attention-weighted fusion.

This design provides complementary information that is not directly available from a purely image-based CNN or transformer representation. The ablation results support this observation: incorporating the morphology branch increases accuracy from 75.26% to 97.65%, while attention-weighted fusion further improves the result to 98.02%. The proposed approach consequently offers a balance between competitive classification performance, efficient feature extraction, and interpretable morphological information.

TABLE IV: Per-magnification performance of the attention-weighted model.

Magnification	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
40X	97.84	97.5	98.0	97.7
100X	98.12	97.9	98.4	98.1
200X	97.91	97.6	98.1	97.8
400X	98.02	97.8	98.3	98.0
Overall (mean)	97.97	97.7	98.2	97.9

TABLE V: Ablation study on the BreakHis test set.

Setting	Macenko	Morph.	Attn.	Acc. (%)
CNN only	✓	–	–	75.26
CNN + Morph. (concat)	✓	✓	–	97.65
CNN + Morph. (attention)	✓	✓	✓	98.02
CNN + Morph. (separate)	✓	✓	(sep.)	97.45
No stain normalization	–	✓	✓	82.36

V. Discussion

The experimental results demonstrate that combining learned visual representations with explicit morphological information can provide a strong and interpretable solution for multi-class breast cancer histopathology classification. The proposed attention-weighted framework achieves 98.02% accuracy, 97.8% precision, 98.3% recall, and 98.0% F1-score on the BreakHis test set. Importantly, the performance is consistent across magnification levels, with accuracies ranging from 97.84% to 98.12%, indicating that the proposed representation is not strongly dependent on a particular acquisition scale. Five-fold cross-validation further produces $98.00\% \pm 0.20\%$ accuracy, supporting the stability of the model across different data partitions.

The ablation results provide direct evidence that the morphological branch contributes substantial complementary information. The CNN-only configuration achieves 75.26% accuracy, whereas adding the seven morphological descriptors increases accuracy to 97.65%, corresponding to a gain of 22.39 percentage points. This improvement is considerably larger than the 0.37-point gain obtained by replacing simple concatenation with attention-weighted fusion. The result suggests that the primary benefit comes from combining heterogeneous visual and morphological evidence rather than from increasing the complexity of the fusion mechanism alone. This observation is consistent with the findings of Joseph et al. [14], who demonstrated that handcrafted Hu moments, Haralick texture descriptors, and color histograms can provide useful discriminative information for BreakHis classification. However, the proposed framework differs by combining explicit descriptors with a learned EfficientNet-B0 representation rather than relying on handcrafted descriptors as the principal representation.

The importance of combining complementary feature representations is also reflected in earlier hybrid approaches. Liew et al. [12] combined DenseNet201 features with XGBoost and reported 97% accuracy for both binary and multi-class BreakHis classification. Maleki et al. [15] similarly inves-

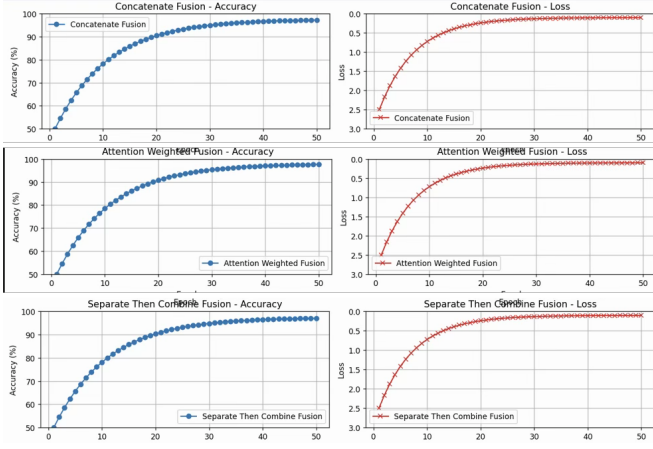


Fig. 6: Training accuracy and loss curves over 50 epochs for the three fusion strategies: (a) concatenation, (b) attention-weighted, (c) separate-then-combine.

TABLE VI: Five-fold cross-validation, attention-weighted model.

Fold	Accuracy (%)	Val. Loss
1	97.85	0.072
2	98.10	0.068
3	97.70	0.078
4	98.25	0.065
5	98.02	0.069
Mean $\pm$ SD	98.00 $\pm$ 0.20	0.070 $\pm$ 0.005

tigated DenseNet201 features with XGBoost, demonstrating the usefulness of separating feature extraction from classification. These studies support the broader principle that learned representations can benefit from complementary processing mechanisms. In the present work, however, the complementary component is not a conventional classifier but an explicit set of seven morphology descriptors, allowing the final representation to contain both high-dimensional learned visual information and low-dimensional, quantitatively defined tissue characteristics.

The results can also be related to the multi-scale strategy of Chattopadhyay et al. [13]. Their MTRRE-Net achieved 97.12%, 95.22%, 96.85%, and 97.81% at 40X, 100X, 200X, and 400X magnifications, respectively, emphasizing the importance of representing information across different tissue scales. The proposed model achieves a narrower performance range of 97.84–98.12% across the same magnification levels. Although the architectures are fundamentally different, this relatively small variation suggests that the combination of stain normalization, learned visual features, and morphology provides a reasonably stable representation across the magnification levels present in BreakHis.

Recent transfer-learning studies further demonstrate the strong performance achievable from pretrained CNNs. Toma et al. [16] evaluated a broad collection of pretrained architectures and reported accuracies as high as 99.8% using ResNeXt-

TABLE VII: Paired t -test results vs. baseline configurations (Bonferroni-corrected $\alpha = 0.0125$).

Comparison	$\Delta\text{Acc. (\%)}$	t	p
Attn. vs. CNN-only	+22.76	31.42	< 0.001
Attn. vs. Concatenation	+0.37	4.81	0.009
Attn. vs. Separate-comb.	+0.57	6.23	0.003
Attn. vs. No stain norm.	+15.66	19.74	< 0.001

TABLE VIII: Comparison with recent prior work on the BreakHis dataset.

Year	Method	Architecture / Strategy	Performance
2021	Oyelade and Ezugwu [11]	Bioinspired NAS-CNN	BreakHis
2021	Liew et al. [12]	DenseNet201 + XGBoost	97%
2022	Chattopadhyay et al. [13]	MTRRE-Net	97.12–97.81%
2022	Joseph et al. [14]	Handcrafted features + DNN	96.10–97.87%
2023	Maleki et al. [15]	DenseNet201 + XGBoost	89.1–93.8%
2023	Toma et al. [16]	Transfer learning CNNs	Up to 99.8%
2024	DeVoe et al. [17]	MobileNetV3	P: 98%, R: 97%
2024	Potsangbam and Devi [18]	DenseNet121 + TL	Up to 96.09%
2025	Yan et al. [19]	DWT + DWNAT-Net	99.66%
2025	Rafiq et al. [20]	DenseNet121	92.50% multi-class
2026	Balaha et al. [21]	Vision Transformer	98.99%
2026	Uddin and Adnan [22]	CaiT + DeiT + Swin	97% multi-class
–	Proposed	EfficientNet-B0 + Morphology	98.02%

50 on BreakHis. Similarly, Potsangbam and Devi [18] reported up to 96.09% using DenseNet121 transfer learning, while DeVoe et al. [17] demonstrated that MobileNetV3 can provide strong classification performance with a lightweight architecture. These studies show that pretrained visual representations remain highly effective for histopathological image classification. The proposed framework should therefore not be interpreted as achieving the highest raw accuracy among all recent BreakHis studies; rather, its contribution lies in augmenting an efficient pretrained CNN representation with explicit morphological evidence and adaptive multimodal fusion.

The increasing use of attention and transformer architectures provides another important comparison. Yan et al. [19] proposed DWNAT-Net, combining discrete wavelet transform with a Neighborhood Attention Transformer, and reported 99.66% accuracy on BreakHis. Rafiq et al. [20] reported 92.50% multi-class accuracy using a DenseNet121-based framework, while Balaha et al. [21] achieved 98.99% accuracy using a Vision Transformer framework. Uddin and Adnan [22] further investigated an ensemble of CaiT, DeiT, and Swin Transformer models and reported 97% multi-class accuracy on BreakHis. These results demonstrate that increasingly sophisticated transformer and ensemble architectures can achieve very high discriminative performance. However, direct numerical ranking against the proposed model is inappropriate because the studies differ in classification formulation, preprocessing, data partitioning, magnification handling, and reported evaluation protocols. In particular, some reported results correspond

to binary classification rather than the eight-class problem addressed in this work.

A distinguishing characteristic of the proposed framework is therefore the explicit integration of morphology with deep visual representations. Transformer-based and CNN-based systems can learn complex tissue patterns directly from images, but their learned features are not inherently expressed in terms of quantitative descriptors such as area, perimeter, eccentricity, solidity, extent, intensity, and compactness. In contrast, the proposed morphology branch provides a compact representation whose variables have direct geometric or intensity-based interpretations. The attention-weighted fusion mechanism then allows the relative contribution of the 512-dimensional visual representation and the seven-dimensional morphological representation to vary between samples. This provides a different form of interpretability from approaches that rely primarily on post-hoc visualization of learned feature maps.

The statistical analysis further strengthens the comparison. Attention-weighted fusion improves over simple concatenation by 0.37 percentage points and over separate-then-combine fusion by 0.57 percentage points. Although these differences are modest in absolute terms, both are statistically significant after Bonferroni correction, with $p = 0.009$ and $p = 0.003$, respectively. The comparison with the CNN-only configuration is substantially larger, with a 22.76-point improvement and $p < 0.001$. Similarly, removing stain normalization reduces accuracy by 15.66 points, confirming that stain normalization is an important component of the complete pipeline. These results indicate that the proposed gains are not solely attributable to a favorable single train/test split and provide controlled evidence for the contribution of the individual components.

From a practical perspective, the proposed framework also provides a balance between model complexity and interpretability. Recent transformer ensembles such as TransBreastNet [22] demonstrate the value of combining multiple learned representations, but ensemble architectures can introduce additional computational and deployment complexity. The proposed method instead uses a single EfficientNet-B0 backbone together with a small seven-dimensional morphological branch. Thus, the framework retains the efficiency advantages of a compact CNN while introducing an explicit source of domain-informed information. This design may be useful in settings where classification performance, computational practicality, and interpretable evidence are all important considerations.

Several limitations should nevertheless be acknowledged. First, the experiments are based on BreakHis, which contains images from a limited number of patients and acquisition settings. Consequently, the observed performance does not establish generalization to other institutions, scanners, staining protocols, or patient populations. The substantial 15.66-point reduction observed without stain normalization further indicates sensitivity to staining variation. Second, the morphological descriptors are extracted from an Otsu-segmented tissue mask at the patch level rather than from individual nuclei. Consequently, the morphology branch captures global

or regional tissue characteristics but does not explicitly model nucleus-level atypia, nuclear pleomorphism, or cell-to-cell spatial relationships. Third, although attention weights provide a more direct representation of feature contributions than a purely opaque classifier, their clinical correspondence has not been validated through a formal pathologist reader study. Finally, comparisons with recent literature should be interpreted cautiously because reported BreakHis results vary in task definition, experimental protocol, and evaluation methodology.

Overall, the findings indicate that the value of the proposed framework is not simply its numerical accuracy but the combination of complementary information sources under a controlled experimental design. The results show that explicit morphological descriptors can substantially strengthen an efficient CNN representation, while attention-weighted fusion provides a statistically supported improvement over simpler fusion strategies. In comparison with recent CNN, transfer-learning, and transformer-based approaches, the proposed framework therefore offers a complementary direction that prioritizes the joint objectives of competitive classification performance, computational efficiency, and interpretable morphological evidence.

VI. Conclusion

We presented a hybrid framework for multi-class breast cancer histopathology classification that fuses EfficientNet-B0 visual embeddings with seven interpretable morphological descriptors through an attention-weighted mechanism. On the BreakHis dataset, the approach achieves 98.02% test accuracy, exceeding CNN-only baselines by over 22 percentage points and outperforming the strongest prior ensemble-based method by 0.72 points, while remaining deployable as a single, computationally efficient model on standard clinical workstations. Five-fold cross-validation and paired significance testing confirm that these gains are stable and statistically robust rather than artifacts of a specific data split. Future work should prioritize external validation on multi-institutional cohorts, extension of the morphological feature set to individual-nucleus-level descriptors, incorporation of additional clinical or genomic metadata, and a formal reader study in which pathologists assess the alignment between the model's attention-based explanations and their own diagnostic reasoning.

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